

Spring 2026 University of Delaware

CHEG/ CISC/ ECEG/ MSEG 848

COMPUTING & DATA SCIENCE FOR SOFT MATERIALS INNOVATION

UD Spring 2026: Computing and data science for soft materials innovation & discovery course

Project Title: Stability of Emulsions used in Fabric coatings

W. L. Gore & Associates Inc., Elkton MD 21921

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Motivation: W. L. Gore & Associates, Inc. manufactures barrier materials (e.g. waterproof breathable jackets, protective vents, etc.) that are used in a variety of industrial applications. These materials act as a barrier layer that provide selective transport of species (e.g. provide for motion of air while blocking liquid water from entering). The materials are heterogeneous with a microstructure that is characterized by one or more solid phases and a void phase. To achieve specific performance goals these materials are often coated, with a solvent-based coating that modifies the interfacial chemistry.

The coating solution is typically an emulsion made of a polymer, solvent, and one or more additives, typically surfactants that are used to stabilize the emulsion. The coating process is identical, in principle, to making French toast where the bread is the porous material, the egg-milk mixture is our aqueous based coating solution with the water being evaporated in the cooking process to leave behind caramelized sugar and protein coating.

The **goal of the project is to** use simulations to understand quantitatively the stability of coating solution in a) bulk solution form and b) as a thin film on a fabric surface. wetting and adsorption of surfactant solutions in porous media and understand the influence of chemistry and microstructure on the wetting properties of the media. **Our main question here is what composition ranges of the coating solution are stable in both bulk and thin film forms.** Corollaries to this central question are –

- a) Do the polymer and surfactant in the solution adsorb to the surface?
- b) what structures do they form on the surface? and
- c) how does the underlying surface chemistry affect this adsorption?

These are the expected stages in this project:

Stage 1: Choose an appropriate model for this problem – coarse-grained or atomistic? If you decide on coarse-grained (CG) what type of CG model - generic or Martini-type? Is solvent explicit or implicit? How will you model the bulk solution and a thin film on surface that represents woven fabric with different surface chemistry – Hydrophobic, neutral, and hydrophilic.

Note: Depending on the type of model decided in Stage 1, you will have to consider how the fabric surface will be modeled. For example, if considering atomistic simulations that past literature shows that self-assembled monolayers can be used as a flat surface with end-groups changed to alter the surface chemistry (e.g., <https://doi.org/10.1039/C3SM50958C>). If considering coarse-grained simulations, the flat surface can be modeled strategically with interaction potentials mimicking the three chemistries - hydrophobic, neutral, and hydrophilic. Also, estimate the length and time scales that can be captured with this model and molecular dynamics simulation method. Decide on the simulation engine to be used for this project – Gromacs or LAMMPS.

Stage 2: Select specific polymer – water – surfactant compositions and run bulk solution simulations and calculate measures of polymer and surfactant dispersion. For this project we will consider one of two commercial surfactants – Tween 20 and Tween 80 - whose properties are available in the literature. The end deliverable here will be a (partial) phase diagram of the emulsion to predict composition ranges where the emulsion is stable. **(Deliverables #1)**

Stage 3: Pick compositions from Stage 2 that resulted in stable emulsions and run thin film simulations on surfaces chosen in Stage 1. Some parameters of interest here are the initial thickness of the film, and the substrate surface (weave, chemistry/charge (e.g. Nylon or Polyester)). Gore can help downselect these variables to keep timelines for calculations reasonable **(Deliverables #2)**

Stage 4: Using quantitative metrics describe the stability of the film, and adsorption behavior of the polymer and surfactant on the surface. (e.g., <https://doi.org/10.1039/c9sm02104c> and <https://pubs.acs.org/doi/10.1021/ie7017383>). What conditions do the polymer spread out and adsorb? What drives this process– charges, chemistry? **(Deliverables #3)**

Skills that are needed for tackling this project:

- Working knowledge of molecular modeling and simulations.
- Polymer science and assembly.
- Experience in C++/python programming at a level where you are comfortable working with object-oriented programming concepts, simulation trajectory file formats, and packaging analysis code into reusable libraries.
- Data science and machine learning techniques for regression.

Code and data considerations:

1. The models will be built on top of open-source modeling and ML frameworks such as
 - a. LAMMPS/Gromacs/OpenMM/HOOMD-blue for molecular simulations
 - b. Pytorch and scikit-learn for ML models

2. All code and simulation setup files will be version controlled in a git repository. Trajectory data will not be stored under version control.
3. All code and data are under a permissive license such as creative commons. All code and data generated in this course will fall under the same permissive license.